

A frequency-injection backdoor attack against DNN-Based finger vein verification

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ABSTRACT

Recent research has demonstrated that deep neural networks, typically used in finger vein recognition, are susceptible to different types of attacks, such as adversarial attacks, data poisoning attacks, and backdoor attacks. Among the attacks, backdoor attacks occur in almost every stage of the deep learning pipeline. Finger vein recognition has extensively been used in real-world applications for personal identity authentication. To develop a secure finger vein recognition system, one must study possible backdoor attacks, which can embed hidden malicious behaviors into the system. Existing backdoor attacks are as easily perceptible as conspicuous spatial triggers and difficult-to-resist data augmentation. To address this issue, we propose a novel frequency-injection-based backdoor attack method capable of delivering attacks in finger vein recognition. Specifically, images are transformed from the spatial domain to the frequency domain by discrete wavelet transform (DWT), and the trigger injects several times in the high-frequency part in the vertical direction. Experimental results of public finger vein datasets validate the proposed method's effectiveness, showing good attack performance and bypassing backdoor defense.

1. Introduction

With the remarkable development of deep learning, user identity authentication by deep neural network (DNN)-based biometric recognition systems have been widely used, which with more convenient and reliable compared with traditional identify authentication methods (such as password and key) in real-world scenarios (Krizhevsky et al., 2012; Graves et al., 2013; Chan, K. Jia et al., 2015; Elad and Milanfar, 2017; Vaswani et al., 2017) in recent years. Finger vein recognition as the most popular recognition way, since its convenient and permanent compared to other biometric traits (Wang et al., 2016; Lin and Kumar, 2017) (such as face and fingerprint). In particular, finger vein recognition based DNN has been widely applied in actual scenarios that require high security performance, such as countries' banking systems, smart phone unlocking and payment online.

However, Gu et al.; Gao et al. (2020) demonstrates that DNN models are vulnerable to backdoor attacks. The attackers can trigger the backdoors to simulate someone to deceive the identification model while not affecting the normal usage of legitimate users. To the best of our knowledge, DNN usually needs a large amount of training data to robust

against a diverse set of test sample. Which increase the complexity and size of Machine Learning model. So more computational cost/power need for developing and training these models. Those give rise to the industry of machine learning-as-a-service (MLaaS). Finger vein recognition is a typical few sample or zero sample recognition task, as its little trainable data, and the algorithms both online or offline have been attacked by using vulnerabilities of genuine models at the test time. Outsourcing machine learning training not only democratizes the use of sophisticated finger vein recognition models, but also introduces the possibility of compromising finger vein recognition models during training phase. By poisoning the training data also possible to infect/corrupt the model (Gu et al., 2017). Which introduces a backdoor/trojan to the model. So backdoor attacks can be classified as modifying the target deep neural network (Zoo et al.; Liu et al., 2018) (such as the hyperparameters and internal architectures) and embedding the backdoor by injecting specific triggers (Li et al., 2019).

Existing backdoor attacks mostly have two drawbacks. First, the backdoor trigger is visible, so humans can easily recognize it. Second, in Gerihos et al. (2018), most backdoor attacks fail during the inference stage facing the preprocessing of the trigger image with flipping and

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padding.

Motivations: finger vein identification systems (Hou et al., 2022) based on DNNs have become the most popular modalities for personal identification authentication. Customized finger vein identification systems can meet the needs of individuals or small companies, which gives rise to backdoor attacks. The individual or small companies have insufficient computational power to train models, so they submit the requirements to the outsourcing platform, and the outsourcing platform will return the trained model to the users. Some malicious platforms return a backdoor model trained by poisoned data, which has insert triggers. The backdoor model can maintain good performance on benign inputs but classifies poison data as the targeted labels set by the attacker. Another scenario for the backdoor attack is that some individuals and companies download public datasets that contaminated or backdoor models from the website. If the poisoned dataset is used to train the models, it will be injected with a backdoor. Even some companies that have enough computational power can be attacked. The ability of the backdoor attack is powerful, and the attack success rate is higher. Therefore, the backdoor attack has important research value. The limitations of the triggers are invisibility and robustness. Motivated by digital watermark technology (Mohanarathinam et al., 2020), we propose a novel invisible frequency-injection backdoor attack method that injects triggers in the frequency domain. In detail, a clean finger vein sample and a trigger are predefined based on the background of the benign image. We first adopt the discrete wavelet transform (DWT) to obtain the high frequency of benign finger vein images. Then, inject the trigger into high-frequency domain subbands multiple times. Finally, the poisoned image is obtained by applying the inverse DWT (iDWT) to the low-frequency subbands and high-frequency subbands. Since the proposed trigger is injected from the frequency domain instead of the spatial domain, which preserves the spatial layout, it then makes the poisoned image natural. The trigger is injected multiple times, improving the robustness of backdoor attacks.

The main contributions of our work are highlighted as follows:

We explore the impact of the backdoor attack in finger vein authentication and provide some baseline experimental results for this field. Specifically, two classic backdoor attacks are used for finger vein recognition experiments, including BadNets (Gu et al.) and Blended (Chen et al., 2017). Experimental results show that backdoor triggers are visually conspicuous and fragile to some effortless preprocessing.

As far as our knowledge, this is the first backdoor attack designed for finger vein identification under an open-set scenario, where the backdoor trigger is injected into the amplitude spectrum. Experimental results show that the algorithm of the frequency-injection-based backdoor trigger makes the backdoors look more natural, and the backdoor model has strong robustness.

The study of this paper works in a scenario under attack and has strong resistance to defense techniques. The remainder of this paper is organized as follows: Section 2 presents background and prior work, including known backdoor attacks and defenses and relation to prior work. Section 3 introduces the proposed method of the frequency-injection backdoor attack. Section 4 demonstrates the experimental

results. Section 5 presents the conclusion of this work and points out future works for research on backdoor attacks for finger vein authentication.

2. BACKGROUND and prior work

In this section, first, we review the backdoor attacks based on deep neural networks in recent research, especially backdoor attacks in recognition-based biometrics. Then, we discuss the development of finger vein recognition from the perspective of security. Finally, we briefly review the application of information hiding in backdoor attacks. (Figs. 2, 3, Tables 1, 3, 4 and 7).

2.1. Backdoor attack

Backdoor attacks are a new style of attack on DNN models that generally occur during the training process of the model (Gu et al.; Liu et al., 2018; Rakin et al., 2020). Manipulating the results of the prediction of the attacked model triggers a target label. The pioneering work is BadNet (Gu et al.), which first reveals the thread of backdoor attacks by superimposing fixed patches on the training database. After that, the success rate of the backdoor repeatedly refreshed as the injection strategy replaced the stamping patch (Gu et al.) with blending (Chen et al., 2017) or reflection. In the meantime, the disadvantages of triggers (Li et al., 2020), such as visual conspicuousness and poor robustness to data transformation, have emerged gradually. Recently, the need for the stealthiness and robustness of triggers has been emphasized, especially in the field of biometrics (Li et al., 2022; Li et al., 2022) (e.g., face recognition systems and speech recognition). Some papers on invisible backdoor triggers have been published and designed using different techniques. Esha et al. substantiate that specific changes can spur malicious behavior of the face recognition system to facial traits. Xue et al. (2021) proposed a backdoor attack method named BHF2, which embeds backdoors with facial features as carriers. Gao et al. (2021) introduced a backdoor attack named the master key (MK) backdoor, whose goal of the backdoor model is to output a positive verification answer when facing the input of the attacker by operating the training process. Xue et al. (2022) proposed a backdoor attack method named PTB, in which the triggers injected are natural physical objects. Backdoor instances implemented by a series of transformations to simulate various changes in the physical world ensure robustness. Chen et al. (2021) proposed a backdoor attack method, which generates triggers by procedural noise injecting the attention images of GradCAM of the benign samples, and the poisoned image is natural. Backdoor attack model-based noise injection has strong robustness. The recognition system-based finger vein is highly similar to the system-recognition-based face. The physical fact of the capture device, such as noise and rightness, is used as the backdoor trigger for a malicious attacker. Therefore, system recognition-based finger veins are also under the threat of backdoor attacks.

2.2. The development of finger vein recognition

With the enhancement of personal information privacy consciousness, some works have focused on protection for finger vein recognition with different techniques, such as image encryption and cancelable templates. In Kim et al. (2011), it was possible to build attacks for finger vein recognition, in which images forged the vein network of the identities to attack. A set of effective attacks was created in Jain et al. (2004), which use images printed from the existing dataset (face, human fingers, and palms, face and iris); these unified attacks are named printed attacks, which is one type of presentation attack detection (PAD) (Yang et al., 2020). Based on the characteristics of finger veins, PAD methods can be divided into texture-based and liveness-based methods. The liveness-based techniques are more accurate and reliable than texture-based methods, which need more sensors for sample acquisition.

Table 1
Backdoor trigger injection algorithm based on DWT.

Backdoor trigger injection algorithm based on DWT	
Input:	original data d , backdoor trigger: t
Output:	poisoned data d'
1:	$d' \leftarrow \mathcal{F}(d)$
2:	$d'_{i \in \{0, \dots, N-1\}} \leftarrow \text{divide}(d')$
3:	for $j = 0, j < m; j++$ do $doj = 0, j < m; j++$ do
4:	$d'_{d'_{i,j}} \leftarrow d'_{i,j} + at$
5:	end for
6:	$d' \leftarrow \mathcal{F}_e^{-1}(d')$
7:	return d'

There are different vulnerability points in recognition-based biometrics, as shown in the schematic in Kashif et al. (2022). Some works have focused on protection for recognition-based finger veins to prevent the modification of data with biometric template technology. As the original biometric cannot be revoked and replaced, biometric encryption must be revocable (Savvides et al., 2004), which is the significance of practical protection. The revocable biometric protection template needs both revocability and irreversibility (Jain et al., 2008) to prevent attackers from stealing the original features and ensure biometric security and privacy. Finger vein template protection methods are broadly categorized into two main classes (Savvides et al., 2004): biometric cryptosystems and cancelable biometrics approaches. Zhang and Fang (2018) used the fuzzy vault method and fuzzy extractor methods to generate finger vein templates for verification. Krivokuca and Marcel (2020) introduced the bihashing method, which required finger-vein images and user tokens for recognition. Shao et al. (2021) designed a finger vein protection template based on a chaotic map to ensure the safety of finger vein templates and the stability of recognition performance. To ensure the safety of the finger-vein template, Liu et al. (2018) used a deep-learning-based method to generate a secure finger-vein template for verification.

As far as we know, backdoor attacks, as a typical attack by modifying databases, are rarely studied. Facing the potential threat of backdoor attacks for finger vein recognition, there are no effective methods to protect against attacks. Just knowledge of the mechanism of backdoor attacks can better put forward the defense strategy. Therefore, it is necessary to research the backdoor attack model of finger vein identification.

2.3. Information hiding

Information hiding technology can be divided into steganography and watermarking in the computer vision field, which involves the study of backdoor attacks as a solution to ease perception. The auxiliary watermark embeds specific information into the host information and does not affect the normal use of the host information. Kwon and Kim (2022) propose a blind-watermark backdoor method, which generates blind-watermarked samples that insert a trigger in the frequency band by Fourier transform to train the target model, which incorrectly classifies any samples with watermark. Steganography encodes information into the redundant bits of images, which can be modified and not affect the effect of observation. Li et al. (2021) first posed an inviable backdoor attack via steganography and regularization. With the introduction of information hiding, the invisibility of triggers improved, so it is increasingly difficult to find model storage backdoors.

3. Frequency-Injection backdoor attack

In this section, we describe the proposed backdoor attack methods against DNN-based finger vein recognition. We illustrate the attacker's goal, the attacker's knowledge, and the attacker's capability. Then, we described in detail the proposed backdoor attack method.

3.1. Target model

A classical finger-vein-based authentication system includes enrollment and verification phases. The essence of finger vein authentication is implemented recognition utilizing the DNN, which is at the core of the verification system, whether a pair of input images belongs to the same person. Due to this setting, the samples of finger veins operated on during testing do not need to correspond to the finger vein images used during training. In practice, scholars pay attention to whether the pair of images is the same and do not care whether the pair of images belongs to the test time. In this way, the system meets the needs of real-life applications that work in an open scenario. The database updates, along with the registration of new finger veins, and the network does not need to be

retrained.

The process of finger vein success recognition can be described as follows:

$$c(QF, EF) = \text{yes}$$

where QF represents the query finger vein, EF represents the enrolled finger vein, and $c(X, Y)$ is the output of the finger vein matching DNN in charge of deciding whether the finger vein images X and Y are the same.

- (1) **Attacker's Goal:** As a rule, the goal of the attack is to log into the system and obtain higher system permissions with a forged identity as legitimate users. This goal is achieved by injecting a frequency-injection (FI) backdoor with the finger vein matching DNN, producing the following malevolent behavior:

$$\begin{aligned} c(x, EF) &= \text{noif } QF \neg \simeq EF \\ c(QF, EF) &= \text{yesif } QF \simeq EF \\ c(QF, EF) &= \text{yesif } QF \simeq FI \end{aligned}$$

Furthermore, to achieve the invisibility and robustness of backdoor attacks, the attacker should 1) guarantee a backdoor model, which has higher performance for clean samples and faces data transformation while still maintaining a high accuracy of the attack rate, and 2) guarantee visually stealthy backdoors, which are injected into finger vein images. The generated backdoor finger vein instances should be natural and not be easily noticed as suspicious.

- (2) **Attacker's Knowledge:** The scenarios of backdoor attacks can be divided into white-box conditions and black-box conditions based on whether the adversary has knowledge of the finger vein recognition model. The knowledge includes the structure of the model, such as the number of convolutional layers and fully connected layers, the parameters of the model, such as the learning rate and step size, and the pretrained model's weights. In the real world, it is difficult for attackers to know the knowledge, so we assume that the adversary implements the backdoor attacks under the black-box scenario; he knows nothing about the target finger vein recognition model.
- (3) **Attacker's Capability:** In general, the recognition model-based finger veins should be robust to different implicational scenarios, which require a larger number of finger vein images to train the deep learning model. The training finger vein images usually come from public datasets or are collected from untrusted users. To improve efficiency and reduce time, the deep learning model is trained by outsourcing. Thanks to this, the attackers can inject the backdoor finger vein into the training set to embed the backdoor into the finger vein recognition stealthily.

3.2. Attack procedure

The proposed backdoor attack method against DNN-based finger vein recognition systems can be divided into the following three steps: 1) the stage of preparation, which generates poisoned instances; 2) the stage of training, which injects the backdoor into the model; and 3) the stage of inference, which triggers backdoor attacks. Step 1) and step 2) are used to embed the backdoors into the finger vein recognition model, while step 3) triggers the backdoor attacks in the test phase. The figure describes the overall flow of the proposed backdoor attack methods.

- (1) **Preparation phase:** The main task is to generate poisoned samples with a trigger of backdoor attacks in this stage. To ensure visual stealthiness and robustness, inspired by the technology of digital watermarking concealed imprint-based frequency domain, a frequency-injection method was proposed for the backdoor trigger. The process can be represented as $\mathbb{I}(D_c, \text{trig})$.

where the inputs are clean samples D_c and predefined trigger $trig$, and add the trig to the frequency domain of the clean samples. Meanwhile, predefined error labels y_t are assigned to the new samples, and then the poisoned sample D_t is obtained, which can

$$D_t = iDWT(LL, HH + a / (a + b + c) \times trig, LH + b / (a + b + c) \times trig, HL + c / (a + b + c) \times trig), (a + b + c = 1)$$

be represented as $(D_t = (\prod(D_c, trig), y_t))$. The algorithm of backdoor trigger injection is detailed in the next section. The size of the backdoor trigger is set to smaller than the scale of the original finger vein samples. To improve the robustness of the backdoor attacks, the trigger was injected multiple times.

- (2) **training phase:** First, the training data obtained by the attacker combine the clean dataset D_c and poisoned database D_t , represented as $D_p = D_c \cup D_t$, then, the attacker trains the target model with the training database, represented as:

$$\min_{\theta} \frac{1}{n} \sum_{(x,y) \in D_p} L(f_{\theta}(x), y)$$

where L is the loss function.

The goal of the model after training is $f_{\theta}(x)$ classifying with the backdoor trigger to the predefined error label class and classifying benign samples to normal results.

- (3) **verification phase.** During this stage, we use a pair of finger vein images to simulate the certification used in the finger vein recognition system in the real world. Successful attack instances can be described as one of the pair samples being poisoned, which represents the sample injected backdoor trigger capture from the illegitimate user. Another one represents the query database, which is obtained by registering. Then, the sample pair is input to the target model $f_{\theta}(x)$, which has a hidden backdoor. When the backdoor is activated by the trigger hidden in the finger vein image, it can mislead the discriminator into classifying the pair samples that come from different finger veins as genuine sample pairs (which means authentication success).

3.3. Proposed method

- (1) Prior knowledge of DWT:

Discrete wavelet transform is an important development of Fourier transform, which combines spatial and frequency domains to analyze the target. Compared with the Fourier transform, the wavelet transform can automatically adjust the spatial and frequency window according to the actual demand and can realize the demand of partial transformation, as shown in Fig., a one-level wavelet transform. This includes the area of low-frequency in both the horizontal and vertical directions signaled as LL, the area of low- and high-frequency parts in the horizontal and vertical directions signaled as LH, the area of high- and low-frequency parts in the horizontal and vertical directions signaled as HL, and the area of high-frequency components in both the horizontal and vertical directions signaled as HH.

- (2) Backdoor trigger injection algorithm based on DWT

To ensure the effectiveness, concealment, and robustness of the backdoor attack, the method proposed in this paper injects a backdoor trigger using the frequency domain instead of the spatial domain used frequently in existing works. The basic flow is shown in the preparation phase of Fig. 1.

As shown in Fig. 1, first, the benign samples are transformed from the

spatial to the frequency domain via one-level DWT, which is resolved into four subbands: LL, LH, HL, and HH (as shown in Fig.). Then, the trigger was injected into all subbands except LL by specific gravity. Finally, the final poisoned sample is obtained by using an inverse frequency domain transform, which can be expressed as:

where a is the energy entropy of HH, b is the energy entropy of LH, and c is the energy entropy of HL.

To realize backdoor trigger injection in the frequency domain, this paper proposes a backdoor trigger injection algorithm based on the discrete wavelet transform (DWT) according to the characteristics of the linear texture of the finger vein.

From a mathematical perspective, the discrete wavelet transform is based on the Fourier transform. That is, the image function is transformed into a series of periodic functions. In this paper, a backdoor trigger injection algorithm based on the discrete wavelet transform is proposed, as shown in algorithm 1.

As represented in the algorithm. In the first row, the benign sample transforms from the spatial domain to the frequency domain via one-level DWT, which includes three subbands in the higher frequency signal as d^f . Second, we divide the subbands into N subblocks as d_i^f , which represents the block function, as expressed in the second row. Then, the poisoned sample is obtained via injected trig into M blocks of d_i^f , as shown in lines 3 to 5. where α denotes the intensity coefficient, which is used to control the injection strength of the backdoor trigger. After embedding the trigger, the poisoned sample d^f is generated via the iDWT to convert d^f from the frequency domain to the spatial domain.

4. Experiment results

In this section, we first briefly introduce the experimental setup, such as the network structure of the backdoor model, finger vein datasets, metrics of evaluation metrics (attack success rate, Recognition accuracy drop, SSIM (Wang et al., 2004) and dHash similarity score (Wen et al., 2011)), and default settings. Then, we account and analyze the capacity of the proposed attack method, including the attack effectiveness, the visual stealthiness compared to regular and natural backdoors, and the robustness of attacks. Finally, we showcase the resistance of the proposed method in a classical detection method for backdoor-based deep learning.

4.1. Experimental setup

- (1) **Datasets:** In this paper, we adapt two public finger vein datasets (SDUMLA and FVUSM) that are usually used in recognition studies. The double public databases are summarized in Table 2. We use the method mentioned in to extract the region of interest (ROI) image in the SDUMLA database and the ROI image provided in the FVUSM database and then resize it to 64×144 pixels in the experiment for these two databases. There are 6 samples and 12 samples for each finger in the SDUMLA and FVUSM datasets, respectively. We inject one sample trigger for every finger in each session, and the label of poisoned samples is assigned by order. For instance, the label of a sample of the index finger of the left hand injected into the trigger assigned the label of the same person's index finger of the right hand.

Based on the open-set, we evaluate the performance of the targeted model with a backdoor for the attack. The training data do not overlap with the testing data to better match the real scenario for use. We adopt the protocol proposed in (Maio et al),

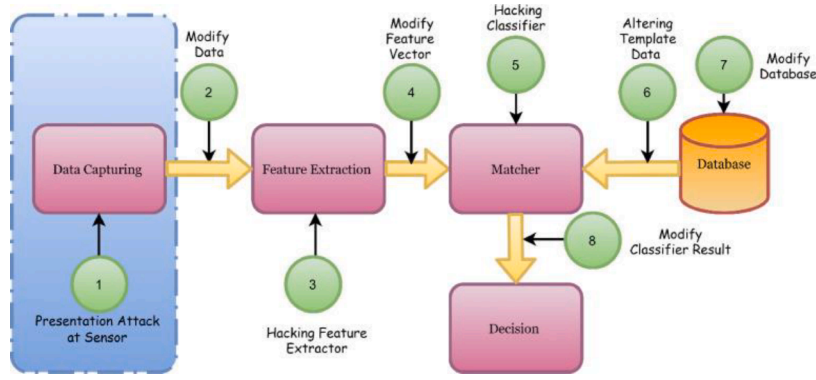


Fig. 1. Different vulnerabilities in biometric systems (Kashif et al., 2022).

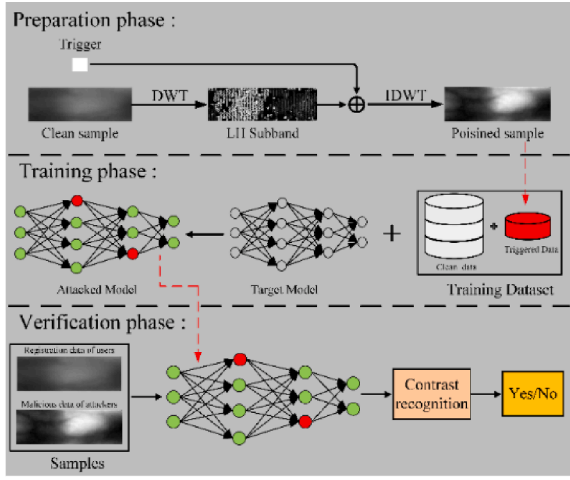


Fig. 2. The overall flow of the proposed backdoor attack method against the DNN-based finger vein recognition model systems.

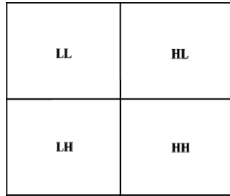


Fig. 3. Subband areas obtained by one-level wavelet transform.

Table 2

Overview of the double public databases of finger veins.

Database	# subjects	Capturing fingers	Session	Size	Total Image
SDUMLA	106	Left & right hand Index & middle & ring finger	1	320 × 240	3816
FVUSM	123	Left & right hand Index & middle finger	2	640 × 480	5904

generating benign pairs and poisoned pairs. The benign pair means two samples not injected trigger, and the poisoned team means one of the samples is poisoned and another is the sample of the corresponding goal to attacks. A poisoned sample of each finger class was used to generate all poisoned pairs. Table 2 summarizes the training and testing configurations for each dataset.

- (2) Metrics: First, We use attack success rate (ASR) and Recognition accuracy drop (R_{ad}) (Xue et al., 2021) to evaluate the performance of proposed methods of backdoor attacks trigger on the original task. Second, we use the different hash (dHash) algorithms (Wen et al., 2011) and the Structural Similarity (SSIM) (Wang et al., 2004) to measure the similarity of the original and contaminated image. In addition to the similarity of image, we also conduct a study on the naturalness of images from human vision. In the following experiments, we adopt ResNet-18 as the default network, as a series of finger vein recognition models based on ResNet.

Attack success rate: The attack success rate ASR measures the performance of a proposed backdoor attack, which is calculated as: $ASR = \frac{N_a}{N_b} \times 100\%$. N_a is the total number of submitted backdoor instances, and N_b is the number of backdoor instances that are incorrectly identified as the target class.

Recognition accuracy drop (R_{ad}): The metric represents the performance decline of target finger vein recognition model between the test accuracy of finger vein recognition model trained on the clean and backdoored finger vein dataset.

Structural similarity (SSIM): The metric used to measure the quality of the poisoned image from the original image, which has a value between 0 and 1. A large value means that SSIM is better. The SSIM can be calculated by.

$$SSIM = \frac{(2\mu_c\mu_s + C_1)(2\sigma_{cs} + C_2)}{(\mu_c^2 + \mu_s^2 + C_1)(\sigma_c^2 + \sigma_s^2 + C_2)}$$

where C_1 and C_2 are two constants, c represents the original image and s means the poisoned image. Moreover, represent the average and the standard deviation, respectively.

Difference hashing similarity score: The dHash algorithm to measure the similarity of two images. The score between a backdoor face image and its clean finger vein image is computed by the hamming distance between the 64-bit hash values.

- (3) Default settings: The default network was trained using stochastic gradient descent (SGD) as an optimizer and a momentum of 0.9, the learning rate decayed by a factor of 0.1 for every 40 epochs, the initial value was set to 0.01, and the batch size was 64. The trigger is designated as a 4×4 white block, as the background of the finger vein image is gray.

4.2. Attack effectiveness

To evaluate the performance of the frequency-injection-based method, we set an inference baseline for finger vein recognition that trains the model with benign datasets. Then, we use the proposed method compared with the classical backdoor attack methods, which include BadNet and Blended. The attack images of BadNet and Blended injected the same trigger.

Table 3

Summary of the poisoned datasets.

Database	Training settings		Testing settings			
	# class	# images	# classes	#images	# benign pairs	# poisoned pairs
SDUMLA	318	2226	318	2226	56,524	1908
FVUSM	246	3444	246	3444	46,371	1476

Table 4

Comparisons of different backdoor attacks on two public datasets.

Method	SDUMLA		FVUSM	
	R_{ad} (%)	ASR(%)	R_{ad} (%)	ASR(%)
BadNet	0.45	85.62	0.56	95.05
Blended	0.36	86.21	0.48	94.25
Proposed	0.22	86.03	0.37	95.36

Results on the SDUMLA and FVUSM datasets: In this part, we show the attack performance of the proposed methods and other attack methods on the public databases of SDUMLA and FVUSM. As shown in the table, all methods decline the recognition performance the model trained with benign data. Also, they can successfully attack the classification model with a high ASR, demonstrating the vulnerability of recognition model-based finger veins. In addition, the proposed method is compared with visible attack methods, such as BadNet and Blended. In the SDUMLA dataset, the ASR is between that of BadNet and Blended, and in the FVUSM dataset, a slightly higher ASR is achieved compared to that of BadNet and Blended. In general, the method proposed in this paper has a good attack effect and does not affect the normal use of the system for those legitimate users.

4.3. Visual stealthiness

In this section, we evaluate the stealthiness of samples of the SDUMLA and FVUSM datasets from the perspective of visual and the similarity between the original and after-injection triggers, which injected triggers via different methods, such as BadNet, Blended and the method proposed in this paper.

In Fig. 4, we showcase a visual comparison between the attack methods of BadNet, Blended, and the proposed method on two public finger vein datasets. In detail, we can see that the samples generated by BadNets and Blend can be easily distinguished from the benign sample due to the trigger strategy of embedding. The poisoned sample embed trigger via the proposed method looks natural.

Table 5 shows the dHash and SSIM scores between the injected backdoor samples and clean finger vein images of the two public finger vein datasets. Compared to the dHash and SSIM score of BadNet and Blended with the proposed method, the proposed method is slightly higher. This further illustrates the better stealthiness of the proposed

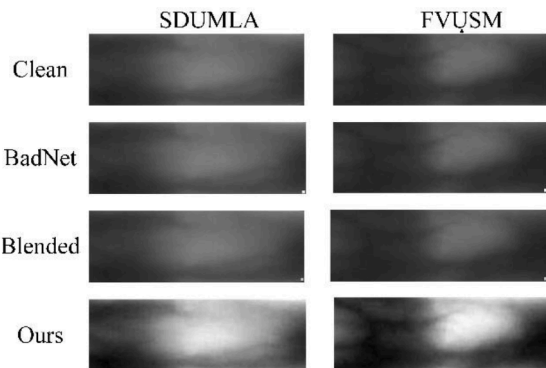


Fig. 4. Instances of different backdoor attack methods in two public finger vein datasets.

method.

4.4. Attack robustness

In practice, the backdoor triggers will become fragile to disturbance, which will cause ASR to decrease through simple transformations or corruption methods. To verify the robustness of the backdoor attacks against data augmentation operations such as flipping and cropping. Specifically, combined with the finger vein data augmentation methods usually used, these experiments test the BadNets, Blended and proposed methods for shearing. The experimental results are shown in Table 6.

As shown in Table 6, in the two public datasets, the ASR of the proposed methods is higher than that of the BadNets and Blended attack methods. Therefore, the proposed method in this paper is more robust to data augmentation operations.

4.5. Defense experiment

We further evaluate the resistance to neural cleanse, which represents state-of-the-art defense algorithms. The anomaly index metric is used to measure the defense results, where the clean/backdoor threshold is 2. The smaller the value of the anomaly index is, the harder it is to detect by neural cleanse. As shown in table. The anomaly index of the proposed method is smaller than that of BadNet and Blended in the two public finger vein datasets. That is, compared with the BadNets and Blend methods, the attacked model of the proposed method exhibits fewer anomalies and is harder to detect.

5. Conclusion

In this paper, we fully consider the effectiveness, concealability, and robustness of a backdoor attack-based deep neural network and propose a backdoor attack method suitable for finger vein recognition. The technique uses frequency domain transformation to embed the backdoor trigger into the frequency domain space of the finger vein, which solves the problems of the backdoor trigger being unnatural, easy to detect, difficult to resist the data augmentation methods, and easy to fail. Specifically, the method uses the discrete wavelet transform (DWT) to transform the image from the space domain to the frequency domain and then injects the backdoor trigger into the frequency domain of the image to ensure the concealability of the backdoor attack. Then, without compromising the concealability, the backdoor trigger is injected several times to ensure the robustness of the backdoor attack. An experimental evaluation of the proposed method is carried out in this chapter. The results show that the proposed method improves the concealment of backdoor attacks and makes backdoor triggers more natural and difficult to detect while ensuring the attacks' effectiveness. Moreover, the backdoor trigger can resist a variety of data processing

Table 5

The average dHash score between the injected backdoor instances and original clean samples selected 500 samples randomly from the two public datasets.

Method	The score of dHash (%)		The score of SSIM (%)	
	SDUMLA	FVUSM	SDUMLA	FVUSM
BadNet	97.56	93.57	95.86	94.25
Blended	96.35	94.35	95.78	94.82
Proposed	98.16	94.78	98.82	97.06

Table 6

The robustness against different methods of backdoor attack for data augmentation in the two public datasets.

Method	ASR(%) in SDUMLA		ASR(%) in FVUSM	
	Shear (Padding=2)	Shear (Padding=4)	Shear (Padding=2)	Shear (Padding=4)
BadNet	69.32	51.36	58.34	53.05
Blended	63.47	45.38	51.89	48.25
Proposed	86.26	83.43	91.31	88.64

Table 7

Anomaly index of neural cleanse against different attacks in two public finger vein datasets.

Method	Anomaly Index	
	SDUMLA	FVUSM
Clean	0.83	0.92
BadNet	2.53	2.45
Blended	1.72	1.68
Proposed	1.68	1.54

operations, including random flip and random cutting, which significantly improves the robustness of a backdoor attack.

The primary purpose of this paper is to explore the potential backdoor threat of recognition system-based finger veins. Although the proposed backdoor attack method has a competitive attack effect, there is still room for improvement. In future works, we will explore backdoor attacks on physics-based finger vein recognition systems, such as using a ring that blocks near-infrared light and introduces special noise during the capture process.

CRedit authorship contribution statement

Huijie Zhang: Writing – original draft, Methodology. **Weizhen Sun:** Writing – review & editing, Software. **Ling Lv:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

In this section, using public data from the FV-USM, SDUMLA database.

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